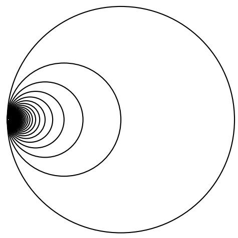
Post-Midsem Homework Problems

4.2 Identification Spaces

1. Which space do we obtain if we take a Möbius strip and identify its boundary circle to a point?
2. Let $f : X \to Y$ be an identification map, let A be a subspace of X, and give $f(A)$ the induced topology from Y. Show that the restrietion $f|_A : A \to f(A)$ need not be an identification map.
3. With the terminology of Problem 3, show that if A is open in X and if f takes open sets to open sets, or if A is closed in X and f takes closed sets to closed sets, then $f|_A : A \to f(A)$ is an identifieation map.
4. Let X denote the union of the circles $[x-(1/n)]^2 + y^2 = (1/n)^2,\ n = 1,2,3,\cdots$ with the subspace topology from the plane, and let Y denote the identifieation space obtained from the real line by identifying all the integers to a single point. Show that X and Y are not homeomorphie. (X is called the *Hawaiian earring*.)



5. Give an example of an identifieation map which is neither open nor closed.
6. Describe each of the following spaces:
   (a) the cylinder with each of its boundary circles identified to a point;
   (b) the torus with the sub set consisting of a meridianal and a longitudinal eircle identified to a point;
   (c) The S^2 with the equator identified to a point;
   (d) The $\mathbb{E}^2$ with each of the circles centre the origin and of integer radius identified to a point.
7. Let X be a compaet Hausdorff space. Show that the cone on X is homeomorphie to the one-point compactification of $X \times [0,1)$. If A is closed in X, show that X/A is homeomorphic to the one-point compactification of $X \setminus A$.
8. Let $f : X \to X'$ be a continuous function and suppose we have partitions $\mathscr{P}, \mathscr{P}'$ of X and X' respectively, such that if two points of X lie in the same member of $\mathscr{P}$, their images under f lie in the same member of $\mathscr{P}'$. If Y, Y' are the identification spaces given by these partitions, show that f induces a map $\hat{f} : Y \to Y'$, and that if f is an identification map then so is $\hat{f}$.
9. Let S^2 be the unit sphere in $\mathbb{E}^3$ and define $f : S^2 \to \mathbb{E}^4$ by $f(x,y,z) = (x^2-y^2, xy, xz, yz)$. Show that f induces an embedding of the projective plane in $\mathbb{E}^4$ (embeddings were defined in Problem 14 of Chapter 3).

10. Show that the function $f : [0, 2\pi] \times [0, \pi] \to \mathbb{E}^5$ defined by

$$f(x, y) = (\cos x, \cos 2y, \sin 2y, \sin x \cos y, \sin x \sin y)$$

induces an embedding of the Klein bottle in $\mathbb{E}^5$.

11. With the notation of Problem 11, show that if $(2 + \cos x)\cos 2y = (2 + \cos x')\cos 2y'$ and $(2+\cos x)\sin 2y = (2+\cos x')\sin 2y'$, then $\cos x = \cos x', \cos 2y = \cos 2y'$, and $\sin 2y = \sin 2y'$. Deduce that the function $g : [0, 2\pi]x[0, \pi] \to \mathbb{E}^3$ given by $g(x, y) = ((2 + \cos x)\cos 2y, (2 + \cos x)\sin 2y, \sin x \cos y, \sin x \sin y)$ induces an embedding of the Klein bottle in $\mathbb{E}^4$.

4.3 Topological Groups

1. Let G be a compact Hausdorff spaee which has the struecure of a group. Show that G is a topological group if the multiplication function $m : G \times G \to G$ is continuous.

2. Prove that every nontrivial discrete subgroup of $\mathbb{R}$ is infinite cycIic.

3. Prove that every non trivial discrete subgroup of the circle is finite and cyclic.

4. Let $A, B \in O(2)$ and suppose $detA = +1$, $detB = -1$. Show that $B^2 = I$ and $BAB^{-1} = A{-}1$. Deduce that every discrete subgroup of O(2) is either cyclic or dihedral.

4.4 Orbit spaces

1. The stabilizer of a point $x \in X$ consists of those elements $g \in G$ for which $g(x) = x$. Show that the stabilizer of any point is a c1osed subgroup of G when X is Hausdorff, and that points in the same orbit have conjugate stabilizers for any X.

2. If G is compact, X Hausdorff, and if G acts transitively on X, show that X is homeomorphic to the orbit space $G/$(stabilizer of x) for any $x \in X$.

with love,
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